

High-fidelity CFD simulations for the wake characteristics of the NTNU BT1 wind turbine

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ABSTRACT

In this work, we present the results of the wake characteristics of a model-scale horizontal-axis wind turbine (HAWT) obtained from high-fidelity CFD simulations. A thorough verification and validation (V&V) study is performed to quantify the numerical uncertainties in the CFD results. The RANS equations with the $k-\omega$ SST turbulence model are adopted in the present study. The geometry of the HAWT is fully resolved including the blades, hub, nacelle, and tower. The Moving-Grid-Formulation (MVG) approach with a sliding interface technique is leveraged to handle the relative motion between the rotating hub and turbine blades and the stationary tower and nacelle. First, to evaluate the spatial and temporal discretization uncertainties in the predicted wake characteristics, a simulation matrix consisting of three systematically refined grids coupled with three different time increments is established. In all the simulations, a tip speed ratio (TSR) of 6 and an inlet velocity of 10 m/s are specified. The numerical uncertainties for the predicted velocity and turbulent kinetic energy at different downstream locations are then obtained by applying a systematic verification procedure to the CFD results. Second, a validation study is carried out by comparing the CFD results against the experimental data. It is shown that the CFD predictions are in good agreement with the measurements. In addition, the details of the wind turbine wake are visualized and discussed. It is found that the tower wake is skewed by the rotating turbine blades. As the result, the wake profiles at downstream locations are asymmetric. Further, the tower wake is carried upwards by the rotating blade wakes, and thus the location of the asymmetry peak at different downstream distances is changing. We therefore confirm that the asymmetry of the wake profile is physical and it is not a measurement error in the experiment as suspected by other researchers in the previous studies. Finally, the interaction between the rotor wake and the tower wake is visualized and discussed. It is observed that due to the presence of the tower, the wake below the hub height is highly unsteady compared to the wake above the hub height. The results show that the methodologies adopted in this study are capable of accurately capturing the wake characteristics of the NTNU BT1 wind turbine.

1. Introduction

A horizontal-axis wind turbine (HAWT) will affect the airflow around it both in the upstream and downstream directions. The upwind flow region which is influenced by a HAWT is known as the induction region, and it is characterized mainly by a reduction of the wind speed [1]. The airflow downstream of a HAWT is called the turbine wake, and it is characterized by a decreased wind velocity and an increased level of turbulence compared to the free stream condition [2]. The turbine wake can be subdivided into two regions: (1) the near-wake region in which the flow structures can be affected by the turbine

geometries, i.e. the blades, hub, nacelle, and tower; and (2) the far-wake region where the flow characteristics are mainly dominated by the global HAWT parameters, i.e. the thrust and power coefficients [3]. In modern wind farms both onshore and offshore, wind turbines are placed in clusters, and thus the wind turbines in a wind farm will be inevitably operating in the wakes of upstream turbines. This phenomenon leads to a reduction of the power generation efficiency of the downstream wind turbines while increasing the fatigue loading on their blades, and eventually reduces the total power generation of the wind farm [4]. Data shows that the total power loss of a large wind farm is between 10% to 25% [5]. Therefore, to further reduce the leveled

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cost of energy (LCOE) of wind farms and to fulfill the potential of wind energy, more efficient wind farms need to be designed. To achieve this goal, accurate predictions of the wind turbine wake are crucial since it forms the backbone of wind farm layout optimizations.

In the past several years, Computational Fluid Dynamics (CFD) methods with different levels of complexity were developed for wind energy applications and have been extensively used in the predictions of wind turbine wakes. Due to the multi-scale nature of the wind turbine wakes, simultaneously revolving the blade boundary layers and the wake regions is challenging. One approach to circumvent this obstacle is to leverage the blade element method (BEM) and its improved methods [6,7], i.e. the actuator disc (AD) [8,9], and the actuator line (AL) [10,11] methods, to simplify the rotating blades as forces acting in the fluid field. Then, those models are coupled with the Reynolds-Averaged Navier–Stokes (RANS) or Large Eddy Simulation (LES) solvers to calculate the flow behind the turbines [2,12–15]. This approach has also been extensively applied to wind farm simulations [5,16,17]. To account for the effects of the nacelle and the tower, immersed boundary method (IB) is also incorporated into the RANS/LES-BEM simulations [18,19]. However, due to the embedded assumption of the BEM-based methods that the flow on the blade surface is two-dimensional, the credibility of this approach is questioned in real-world applications where the flow in the vicinity of the blades is highly three-dimensional [20,21].

With the continual decrease of computational cost in recent years, wind turbine simulations with fully resolved geometries have become more popular. This approach is preferable in simulations where the flow details in the vicinity of the wind turbines are critical. For example, the airflow around a floating offshore wind turbine (FOWT) needs to be precisely resolved in order to perform an accurate analysis for the aerodynamics of the FOWT in waves and currents [22–25]. Under certain conditions, the blades will even operate in their own wakes due to the platform motions [21,26,27]. In addition, scholars have demonstrated that the effect of the tower and the rotor-tower interaction is not negligible on the wake characteristics of a wind turbine [28–30], thus correctly resolving the rotor and tower geometries is necessary. Gómez and Seume [31] clearly showed the effect of the rotating blades on the tower wake by using a two-dimensional RANS calculation. It was found that due to the rotation of the turbine blades, the tower wake was skewed, and this phenomenon cannot be accounted for by using traditional BEM methods. Later, by adopting fully resolved rotor and tower geometries in the CFD simulations, the interactions between the rotor and tower wakes were further elucidated for both upwind [32,33] and downwind [34–38] types of HAWTs.

In the present study, high-fidelity CFD simulations are performed for the Norwegian University of Science and Technology (NTNU) Blind Test (BT) 1 wind turbine to investigate the wake characteristics at different downstream locations. In prior studies, CFD simulations with different rotor-modeling approaches, i.e. BEM-based methods [19,39–41] and fully resolved geometries [42–45], coupled with different turbulence models have been performed for the BT1 wind turbine, and challenges were identified in the predictions of the asymmetry in the wake profiles [46–49]. Although some scholars have claimed that their methods can reproduce the asymmetric profiles, the underlying mechanism which leads to the asymmetry has not yet been fully understood. In addition, most of those studies only compared the CFD results directly to the measurement, while detailed quantification of the numerical uncertainties was absent or at least insufficient. According to the author's experience, this raised more doubts when considering the fact that plausible results may be obtained by CFD simulations using inaccurate numerical settings. In an effort to address the above concerns in the present study, prior to the examination of the wake characteristics, a systematic verification and validation (V&V) procedure will be applied to the CFD predictions to evaluate the numerical uncertainties of the CFD simulations, and the capabilities and limitations of the methods adopted in the current work will also

be discussed by comparing the current predictions with other numerical results.

This paper is organized as follows. First, the numerical methods and the NTNU BT1 experiment is described. Second, a simulation matrix with 9 cases are established by using 3 systematically refined grids and 3 different time increments. The adopted verification procedure is then applied to the simulation results to estimate the numerical uncertainties in the predicted wake characteristics, and the simulation results are validated against the measurement. Afterwards, the flow details of the simulations are presented and several important aspects of the results, i.e. the skew of the tower wake, the asymmetry in the wake profiles, and the interactions between wakes, are highlighted and discussed in detail. Last, conclusions are drawn from the results and discussions.

2. Methodologies

2.1. CFD approach

The simulations are performed by using MARIN's multi-phase viscous CFD code ReFresco (www.refresco.org). The code solves unsteady incompressible flows using the Navier–Stokes equations. Turbulence closure models based on the Reynolds-Averaged Navier–Stokes (RANS), Detached Eddy Simulation (DES), Partially-Averaged Navier–Stokes (PANS), and Large Eddy Simulation (LES) can be used in ReFresco. The code is parallelized and can run on high-performance computation (HPC) clusters.

2.1.1. Governing equations

An unsteady viscous flow is governed by the Navier–Stokes equations. In the present study, the incompressible Reynolds-Averaged Navier–Stokes (RANS) equations are solved in the simulations:

$$\nabla \cdot \vec{V} = 0 \quad (1)$$

$$\frac{\partial \vec{V}}{\partial t} + \nabla \cdot (\vec{V} \vec{V}) = \nabla \cdot \left[(\nu + \nu_t) (\nabla \vec{V} + (\nabla \vec{V})^T) \right] - \nabla \left(\frac{p}{\rho} + \frac{2}{3} k \right) + \vec{B} \quad (2)$$

where $\vec{V}$ represents the velocity field, t the time, ρ the density, p the pressure, $\vec{B}$ the body force vector, and k the turbulence kinetic energy. ν denotes the kinematic viscosity of air, and ν_t is the eddy viscosity in the Boussinesq hypothesis which is obtained by solving another two transport equations in two-equation turbulence models, i.e. the turbulence kinetic energy k and specific dissipation rate ω in the $k-\omega$ SST model [50]. The two additional transport equations are as follows:

$$\frac{\partial k}{\partial t} + \nabla \cdot (\vec{V} k) = \tilde{P}_k + \nabla \cdot (\Gamma_k \nabla k) - D_k \quad (3)$$

$$\frac{\partial \omega}{\partial t} + \nabla \cdot (\vec{V} \omega) = P_\omega + \nabla \cdot (\Gamma_\omega \nabla \omega) - D_\omega + Y_\omega \quad (4)$$

where $\tilde{P}_k$ and P_ω denote the turbulence production terms, Γ_k and Γ_ω the effective diffusivity of k and ω , respectively. D_k and D_ω are the dissipation terms while Y_ω is the term which blends the standard $k-\omega$ and $k-\epsilon$ turbulence models.

2.1.2. Rotor handling

The rotating wind turbine is handled by leveraging the Moving-Grid-Formulation (MVG) method with the sliding interface technique. For the MVG approach, the integral form of the RANS equations that are applied to a control volume are written as:

$$\int_S (\vec{V} - \vec{V}_g) \cdot \vec{n} dS = 0 \quad (5)$$

$$\int_V \frac{\partial \vec{V}}{\partial t} dV + \int_S \left[\vec{V} (\vec{V} - \vec{V}_g) \cdot \vec{n} \right] dS \quad (6)$$

$$= \int_S (\mathbf{v} + \mathbf{v}_t) \left[(\nabla \bar{\mathbf{v}} + \nabla \bar{\mathbf{v}}^T) \right] \cdot \bar{\mathbf{n}} dS \\ - \int_V \nabla \left(\frac{p}{\rho} + \frac{2}{3} k \right) dV + \int_V \bar{\mathbf{B}} dV$$

where $\bar{\mathbf{v}}_g$ is the motion of grid. In the MVG method, all the variables are defined in the earth-fixed reference system.

2.2. Verification and validation

Systematic verification and validation (V&V) procedures need to be carried out to obtain reliable solutions from CFD simulations. Verification serves to verify the equations we choose are correctly solved, whereas validation serves to indicate whether the appropriate equations are derived to characterize the problem by directly comparing the simulation results to the experimental data.

2.2.1. Numerical errors and uncertainties

The terms “numerical errors” and “numerical uncertainties” are closely related but conceptually different. The “numerical error” is the absolute difference between a numerical prediction and the “exact” solution, whereas the “numerical uncertainty” specifies an interval that contains the “exact” solution with a certain degree of confidence [51]. Typically, the “exact” solution is unknown and thus a method to estimate it from the existing simulation data is required.

Normally, three main types of numerical errors are present in a CFD calculation, which are round-off, iterative, and discretization errors [52]. The total numerical error ϵ_n can be written as:

$$\epsilon_n = \epsilon_{ro} + \epsilon_{it} + \epsilon_d \quad (7)$$

The round-off error ϵ_{ro} is due to the limited floating-point precision of computers, but it is thought to be insignificant when double-precision machines are used. The iterative error ϵ_{it} comes with the fact that the RANS equations are nonlinear, but it is also considered negligible if the solution is well converged. The discretization error ϵ_d , nevertheless, emerges in the process of discretizing the governing partial differential equations into algebraic equations and is considered dominant among the numerical errors in CFD simulations.

The practice to estimate the numerical uncertainty associated with the discretization error is called verification, in which the discretization uncertainty U_d of a numerical prediction ϕ_i is quantified while the “exact” solution ϕ_0 is not known [53].

2.2.2. Verification

The verification procedure adopted in this study is proposed by Eça and Hoekstra [54]. For unsteady simulations, the discretization error can be expressed by a power series as:

$$\epsilon_d \approx \delta_d = \phi_{ij} - \phi_0 = \alpha_x h_i^{p_x} + \alpha_t t_j^{p_t} \quad (8)$$

In the above equation, ϕ_{ij} represents the quantity to be used in the verification study obtained from the simulation using grid i and time increment j . ϕ_0 is the “exact” solution obtained by fitting a set of ϕ_{ij} (e.g. ϕ_{21} , ϕ_{22} , and ϕ_{33}) in a certain manner. p_x and p_t denote the observed orders of convergence in space and time, respectively. h_i is the relative grid size and t_j is the relative time increment. For unstructured grids used in the present study, h_i and t_i are defined as:

$$h_i = \left(\frac{N_{cell1}}{N_{celli}} \right)^{\frac{1}{n_d}} \quad (9)$$

$$t_j = \frac{\Delta t_j}{\Delta t_1} \quad (10)$$

where N_{celli} denotes the number of cells in grid i , and Δt_j the time step size of the time increment j . The subscript 1 represents the finest grid and time increment used in the simulations. Therefore, h_1 and t_1 represent the finest grid spacing and time step size. n_d represents the dimension of the CFD simulations, i.e. $n_d = 3$ in the present study.

There are 5 unknowns in Eq. (8), therefore, at least 5 simulations with different combinations of h_i and t_j are necessary to determine those unknowns. However, Eça and Hoekstra [54] suggested to use more simulations in the error estimation process when using the method of least-squares. Those 5 unknowns are determined by minimizing the following equation:

$$S_{RE}(\phi_0, \alpha_x, p_x, \alpha_t, p_t) = \sqrt{\sum_{i=1}^{n_g} \sum_{j=1}^{n_t} w_{ij} \left[\phi_{ij} - (\phi_0 + \alpha_x h_i^{p_x} + \alpha_t t_j^{p_t}) \right]^2} \quad (11)$$

where n_g and n_t denote the total number of grids and time increments used in the verification study, respectively. w_{ij} represents the weight assigned to a specific prediction such that the relative importance of the predicted results by using different h_i and t_j can be distinguished, e.g., larger weights are assigned to results obtained from finer h_i and t_j . By solving Eq. (11), the “exact” solution ϕ_0 can be estimated, and the standard deviation of the least-squares fit to be used as a metric to quantify the quality of the fitting procedure can be written as:

$$\sigma_d = \sqrt{\frac{\sum_{i=1}^{n_g} \sum_{j=1}^{n_t} n_g n_t w_{ij} \left[\phi_{ij} - (\phi_0 + \alpha_x h_i^{p_x} + \alpha_t t_j^{p_t}) \right]^2}{n_g n_t - 5}} \quad (12)$$

Further, a data range parameter is defined by:

$$\Delta_\phi = \frac{(\phi_{ij})_{\max} - (\phi_{ij})_{\min}}{n_g n_t - 1} \quad (13)$$

where $(\phi_{ij})_{\max}$ and $(\phi_{ij})_{\min}$ are respectively the maximum and minimum values of ϕ_{ij} among all the simulations. The error estimation is considered reliable if $\sigma_d < \Delta_\phi$ is valid, and a safety factor $F_S = 1.25$ will be used. Else, a safety factor $F_S = 3$ will be used. Finally, the discretization uncertainty U_d can be calculated as follows,

For $\sigma_d < \Delta_\phi$,

$$U_d(\phi_{ij}) = F_S \epsilon_d(\phi_{ij}) + \sigma_d + |\phi_{ij} - \phi_{fit}| \quad (14)$$

For $\sigma_d \geq \Delta_\phi$,

$$U_d(\phi_{ij}) = F_S \frac{\epsilon}{\Delta_\phi} [\sigma_d(\phi_{ij}) + \sigma_d + |\phi_{ij} - \phi_{fit}|] \quad (15)$$

where ϕ_{fit} is the value obtained from the fitting procedure corresponding to the grid i and time increment j .

2.2.3. Validation

The validation procedure introduced by [55] is adopted in the present validation study. In the method, the comparison error E_c is defined as:

$$E_c = \frac{\phi_{exp} - \phi_{ij}}{\phi_{exp}} \quad (16)$$

where ϕ_{exp} is the experimentally measured quantity and ϕ_{ij} the simulation predicted value. Eq. (16) represents the absolute difference between the CFD prediction and the experimental measurement.

Then, the validation uncertainty U_v is expressed as:

$$U_v^2 = U_{exp}^2 + U_{sim}^2 + U_{mod}^2 \quad (17)$$

The validation is said to be achieved at the U_v level if $|E_c| < U_v$ is valid. In Eq. (17), U_{exp} denotes the uncertainty of the experimental data and U_{sim} denotes the uncertainties in the numerical predictions which can be estimated from the verification study. U_{mod} is the simulation modeling uncertainty arising from modeling assumptions and it is not practical to be quantified [55], hence will not be considered in the current study. U_{exp} is determined typically by performing repeated experiments. However, this information is not available and thus will not be taken into account in the current validation study.

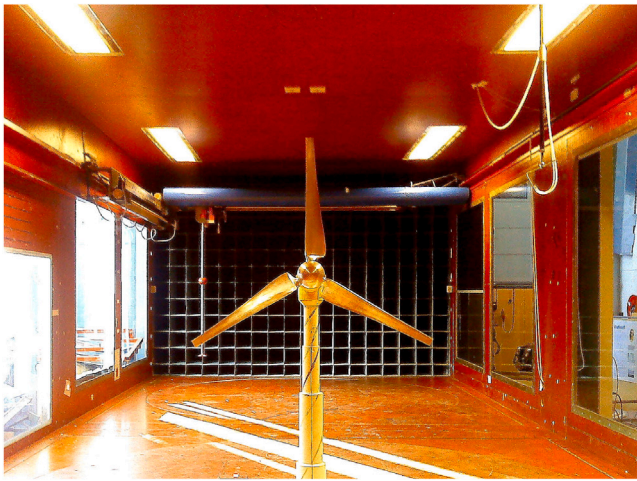


Fig. 1. Experiment setup.
Source: Adopted from [56].

3. Test case description

3.1. NTNU BT1 experiment

In 2008, NTNU started a series of wind tunnel tests with various configurations and input conditions [56]. A horizontal-axis wind turbine constructed by using the NREL S826 airfoil was evaluated in a closed-loop wind tunnel. The total length and width of the wind tunnel are 11.15 m and 2.71 m, respectively, and the height is 1.801 m at the inlet and 1.851 m at the outlet. The rotor diameter is $D = 0.894$ m and the rotor axis is 0.817 m above the ground. The rotor center is placed at the center line of the wind tunnel and is 3.66 m away from the inlet. An illustration of the experiment setup is shown in Fig. 1.

In the experiment, a uniform inflow velocity of 10 m/s and a turbulence intensity level (TI) of 0.3% were used, and the turbine was rotating at a range of tip speed ratios (TSR) from 1 to 12. In the current study, $TSR = 6$ will be used in the CFD simulations and the resulting value of Re is 1.036×10^5 which is calculated based on the chord length of the blades at the tip. This specific TSR value is chosen for two reasons. First, the design TSR of the NTNU BT1 wind turbine is 6, and many previous researchers have performed simulations at this TSR value. Therefore, adopting this TSR value allows our result to be compared with other results. Second, our main focus is on the elucidation of some important mechanisms behind the formation of the asymmetric wake profiles, and thus one single operation condition of the turbine will be sufficient to achieve this goal.

3.2. Computational domain and grid generation

The computational grids are constructed by leveraging the multi-block technique. An illustration of the computational domain is provided in Fig. 2. As shown in the figure, the coordinate system is defined as follows: the origin of the coordinate system is placed at the rotor center, the positive x -axis points from inlet to outlet and y -axis from bottom to top, and the z -axis is thus defined according to the right-hand rule.

The computational domain is decomposed into three separate domains: (1) the inner rotating domain which is a short cylinder containing the rotor; (2) the stationary wake domain which is behind the rotor and extends to the outlet; (3) the stationary outer domain which is generated to match the wind tunnel in the experiment. In the simulations, the inner rotating domain will rotate while the wake and outer domains remain stationary. Those domains are connected by sliding interfaces through which the solution of a domain is extrapolated to pass the flow

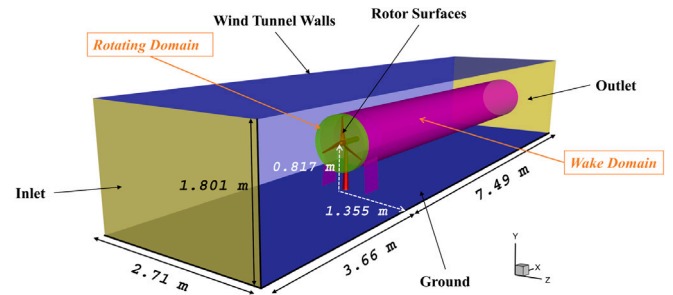


Fig. 2. Computational domain.

Table 1

Numerical settings.

Equations	Preconditioner	Solver	Discr. scheme
Momentum	JACOBI	GMRES	QUICK
Pressure	BJACOBI	CG	—
Turbulence	BJACOBI	GMRES	HARMONIC

information to the neighboring domains. By this decomposition, we can refine an individual part without changing the others. This strategy enables us to perform a systematic verification study specifically for the wake region.

Then, two different grid generation techniques are applied to those three domains: unstructured hexahedral cells are generated for the inner rotating domain which contains the highly twisted blade geometry, while structured grids are generated for the wake and outer domains. For the wind turbine wall boundaries, including the blades, hub, nacelle, and tower, viscous layers are generated with the criterion of $y^+ < 1$, and the cell expansion ratio normal to the wall is 1.2. High refinement levels are applied to the leading and trailing edges and other high-skew regions of the turbine blades. In addition, the meshes at the tip and root regions are further refined to capture the tip and root vortices. Illustrations of the computational grids of the inner rotating domain and the stationary wake domain are shown in Figs. 3 and 4, respectively. It is worth mentioning that in the following verification study, only the grid in the wake domain is refined while the other parts are unchanged. The grid resolution of the inner rotating domain is determined based on the authors' previous studies [49,57,58].

4. V&V study for the wake characteristics

In this section, a thorough V&V study is performed for the predicted wake characteristics of the BT1 wind turbine. A uniform inflow velocity of 10 m/s and a TI of 0.3% are specified at the inlet, and a fixed pressure boundary condition is adopted at the outlet. The no-slip condition is applied to all the wall boundaries. The interfaces between the different domains are set as BCInterface in ReFRESCO, which allows the flow information to be transferred from one to another. Unsteady RANS calculations are performed, and an implicit three time-level scheme [59] is used for the time marching. The choice of the preconditioners, the solvers, and the discretization schemes for the governing equations of the simulations are summarized in Table 1.

4.1. Verification

To quantify the spatial and temporal discretization uncertainties, a simulation matrix consisting of 3 systematically refined computational grids coupled with 3 different time increments is established. It is worth mentioning again that only the computational cells in the wake domain are refined, while the grids in the other domains remain unchanged. Those 3 computational grids are denoted as G1 to G3, in which G1 represents the finest grid while G3 the coarsest. The numbers of cells in

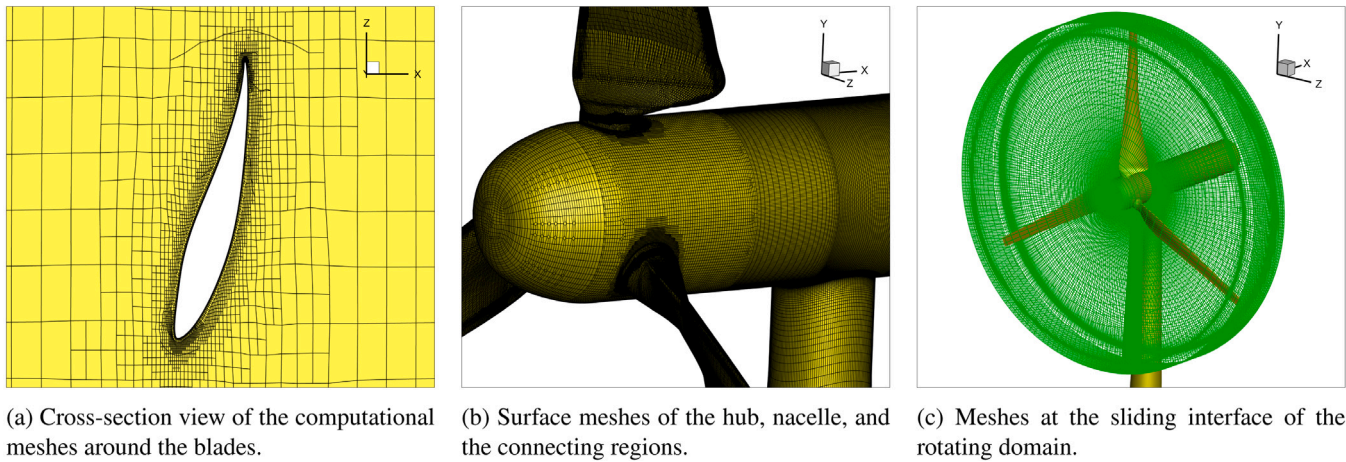


Fig. 3. Illustrations of the computational grid of the inner rotating domain, including the meshes around the blades, the surface meshes of the hub and nacelle, and the meshes at the sliding interfaces.

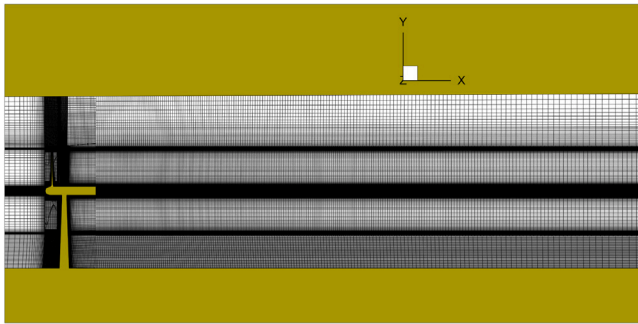


Fig. 4. Illustration of the computational grid of the wake domain.

Table 2

Number of cells in each direction of the computational grids in the wake region.

	Axial	Radial	Circumferential
G1	281	101	360
G2	200	85	256
G3	140	51	180

Table 3

Computational grids and the corresponding relative grid sizes used in the verification study.

Grids	h_i	N_w^i [million]	N_T^i [million]
G1	1.000	10.17	34.15
G2	1.334	4.28	26.96
G3	2.006	1.26	17.46

each direction, i.e. the axial, radial, and circumferential direction of the computational grids in the wake domain is summarized in Table 2. As shown in the table, the number of cells in the circumferential direction ranges from 180 to 360, indicating that the resolution in this direction changes from 2 cells to 1 cell per degree from G3 to G1.

The basic information of the 3 computational grids used in the verification study is summarized in Table 3. h_i denotes the relative grid size and is defined by Eq. (9), N_w^i the number of cells in the wake region, and N_T^i the total number of cells in the computational grids. Note that the meshes in the outer domain is also refined accordingly in the x – $axis$ direction to ensure a smooth transition of the flow information through the sliding interfaces.

The information of the 3 time increments used in the calculations is provided in Table 4. The smallest temporal spacing is denoted as T1 while the largest T3. t_j is the relative time increment defined by

Table 4

Time increments and the corresponding relative time steps used in the verification study.

Time increments	t_j	Δt_j [s]	No. time steps per revolution
T1	1.000	0.0000975	480
T2	1.333	0.00013	360
T3	2.000	0.000195	240

Table 5

Largest theoretical Courant numbers for all 9 simulations.

	G1	G2	G3
T1	0.070	0.050	0.035
T2	0.093	0.066	0.046
T3	0.139	0.099	0.070

Eq. (10), and the real time increment is denoted by Δt_j . The last column is the time steps needed for the wind turbine to rotate a full cycle, e.g., $t_j = 1.0$ corresponds to a 0.75° rotation per time step in the simulations.

4.1.1. Iterative convergence

The iterative convergence of a CFD simulation is represented by the level of residuals. Typically, the residuals are higher in simulations with larger Courant numbers [60]. Here, we define a local Courant number which is calculated based on the cells in the wake domain:

$$Co_w = U_x \frac{\Delta t}{\Delta x_w} \quad (18)$$

where U_x is the streamwise velocity, Δt the time increment, and Δx_w is the minimum cell size of the wake region in the x –direction.

The Courant numbers for the 9 cases are summarized in Table 5. It can be observed that the G1T3 case has the largest Courant number while the G3T1 case has the smallest. The residuals of case G1T3 and case G3T1 are then presented in Fig. 5. Obviously, the L_2 -norm of the residuals for both simulations remained at or under the level of 10^{-4} in the simulations. This indicates that the simulations are well converged and the iterative errors ϵ_{it} in the simulations are small and thus can be legitimately neglected in the estimations of numerical uncertainties.

4.1.2. Discretization uncertainties

The normalized mean wake deficit profiles (referred to as “deficit profiles” hereafter) and the normalized mean turbulence kinetic energy (TKE) profiles (referred to as “TKE profiles” hereafter) obtained from all 9 cases are shown in Figs. 6 and 7, respectively. Note that in the deficit profiles, as shown in Figs. 6(a), 6(c), and 6(e), peaks in the

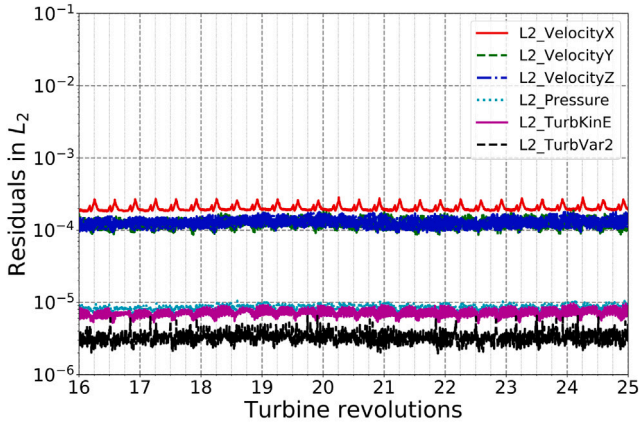
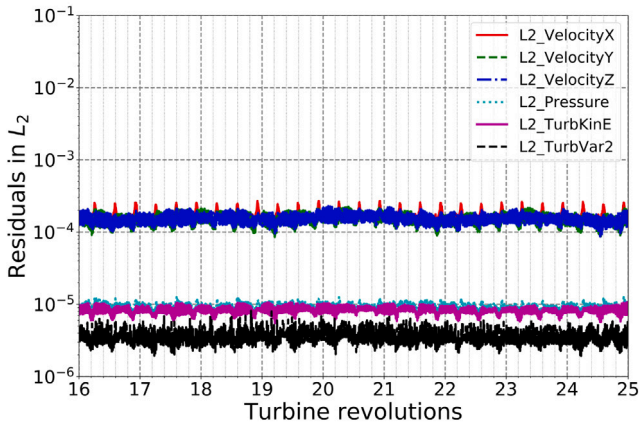
(a) L_2 -norm of the residuals for G1T3.(b) L_2 -norm of the residuals for G3T1.

Fig. 5. L_2 -norm of the residuals for the cases with the largest and the smallest theoretical Courant numbers.

profiles indicate low streamwise velocity values at the corresponding locations. For example in Fig. 6(a), there are two peaks near $z/R = \pm 1.0$, indicating the streamwise velocity is low behind the blade tips (referred to as “tip peaks” hereafter). Furthermore, another peak can be identified around $z/R = -0.3$, as indicated by a yellow box. This third peak emerges due to the existence of the tower wake, causing the wake profiles to be asymmetric about the center, i.e. $z/R = 0$. This peak will be referred to as the “asymmetry peak” hereafter. The mechanism behind this phenomenon will be discussed in detail in the following sections. For the reason that the asymmetry in the deficit profiles is the leading challenge and the primary concern in the simulations of the BT1 wind turbine, in the current verification study, we will choose the values of the asymmetry peaks obtained from the different cases as the quantity to be used in the estimations of the discretization uncertainties. As shown in Figs. 6(a), 6(c), and 6(e), the asymmetry peaks in the deficit profiles at different downstream locations are identified and indicated using yellow boxes, and a zoomed-in view is also provided. The asymmetry peaks in the TKE profiles are also indicated by using yellow boxes, as shown in Figs. 7(a), 7(c), and 7(e). Then, as shown in Figs. 6(b), 6(d), 6(f), 7(b), 7(d), and 7(f), the predicted values of the asymmetry peaks in the deficit and TKE profiles are rendered in two ways for clarity: (1) the results are grouped by the relative time increments and plotted as a function of relative grid size (left); (2) the results are grouped by the relative grid size and plotted as a function of relative time increment (right).

Table 6

Summary of discretization uncertainties of the predicted asymmetry peaks in the deficit profiles.

Loc.	Extrapolated value, ϕ_0	Solution of G1T1, ϕ_{11}	Error ϵ_ϕ [%]	Uncertainty U_ϕ [%]
1D	0.526	0.450	14.4	50.4
3D	0.298	0.354	18.8	29.7
5D	0.343	0.381	11.1	30.7

Table 7

Summary of discretization uncertainties of the predicted asymmetry peaks in the TKE profiles.

Loc.	Extrapolated value, ϕ_0	Solution of G1T1, ϕ_{11}	Error ϵ_ϕ [%]	Uncertainty U_ϕ [%]
1D	1.73×10^{-2}	1.33×10^{-2}	23.1	40.0
3D	3.69×10^{-3}	3.74×10^{-3}	1.4	32.1
5D	2.10×10^{-3}	2.60×10^{-3}	23.8	24.2

Fig. 6 shows the deficit profiles obtained from all cases. In general, at each of the three downstream distances, the profiles obtained from all the 9 cases are asymmetric about the center while being similar to each other. In addition, as shown in Figs. 6(b), 6(d), and 6(f), the values of the asymmetry peaks are more sensitive to the refinement of the grid and time increment at 1 turbine diameter (D) downstream. As the downstream distance further increases, the values of the asymmetry peaks from different cases become close to each other. Fig. 7 shows the TKE profiles obtained from different cases. In accordance with the deficit profiles, the predicted TKE profiles are also asymmetric. The values of the asymmetry peaks of the TKE profiles are shown in Figs. 7(b), 7(d), and 7(f). At 1 D downstream, similar to the deficit profiles, the influence of the refinement of the grid and time increment on the asymmetry peaks is clearly visible. However at 5 D downstream, the predicted values of the asymmetry peaks obtained from all cases are almost identical. This trend is in line with the observation from the deficit profiles.

Then, the verification procedure adopted in this study is applied to the obtained data sets, and the fitting planes are visualized in Fig. 8. The discretization uncertainties (U_ϕ) are estimated for the results obtained by using G1T1 and are summarized in Tables 6 and 7. It can be observed in Table 6 that the maximum estimated value of U_ϕ for the asymmetric peaks appears at 1 D downstream, which is 50.4%. And then, with the increase of the downstream distance, the values of the numerical uncertainty decrease to around 30%. A similar trend is also observed for the U_ϕ of the TKE, as shown in Table 7 in which U_ϕ is 40.0% at 1 D downstream while decreasing to 24.2% at 5 D downstream. Due to the fact that the residuals in the simulations were two orders of magnitudes lower than the discretization uncertainty, the total numerical uncertainty is thus assumed to be contributed solely by the discretization uncertainty. Therefore, the listed uncertainty ranges in Tables 6 and 7 will be adopted in the following validation study.

4.2. Validation

The deficit and TKE profiles obtained from G1T1 are then compared against the measurement, as shown in Fig. 9. The results in the present study are plotted as the red lines, and the experimental data is plotted as the black squares. The uncertainty bars are also indicated by vertical red lines. For comparison purpose, results using other representative modeling approaches reported by Krogstad and Eriksen [56] and Ji et al. [19] are also presented here, i.e. (1) FRG-RANS w/o tower [56] - RANS calculations using fully-resolved rotor geometry but without the tower; (2) FRG-RANS w/ tower [56] - RANS calculations using fully-resolved rotor geometry including the tower; (3) AL-LES [56] - LES simulations in which the rotor is modeled by the actuator line method and no nacelle or tower effect were accounted for; and (4) AL-IB-LES [19] - LES simulations in which the rotor is modeled by the

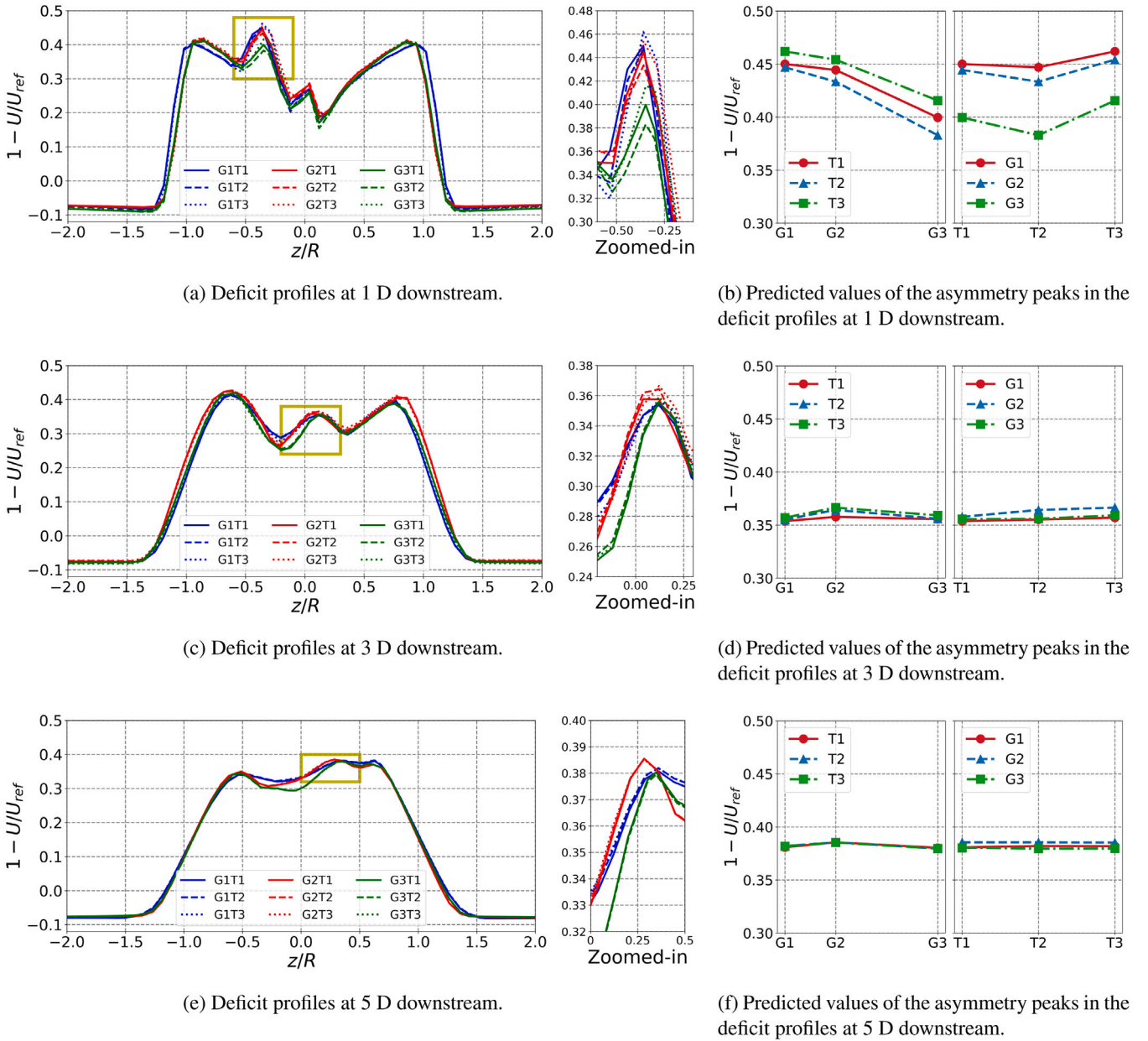


Fig. 6. Deficit profiles obtained from all simulations.

actuator line method and the nacelle effect is modeled by the immersed boundary method.

At 1 D downstream, as shown in Fig. 9(a), the present results successfully captured the asymmetry in the deficit profile, whereas all other results predicted symmetric profiles. Specifically, the present result is the only one captured the asymmetric peak, although the magnitude and the location of it is slightly different from the measurement. For the TKE profiles, as shown in Fig. 9(b), the current result matches the measurement in trend while being roughly 1 order of magnitude lower than it, except for in the range of $-0.5 < z/R < 0.2$. The asymmetry in the TKE profile is also successfully predicted. The AL-IB-LES method also predicted an asymmetric TKE profile and its magnitude almost matches the measurement perfectly. This is interesting because the same method failed to capture the asymmetry in the deficit profile. At 1 D downstream, the asymmetry peaks of the deficit and the TKE profiles measured in the experiment are within the estimated numerical uncertainty ranges, therefore, the numerical results are validated.

At 3 D downstream, as shown in Fig. 9(c), the present results match the measured data well. The asymmetry in the measurement is

captured by the present study while all other numerical results seem to give symmetric profiles. Further, comparing to the deficit profile at 1 D downstream, the location of the asymmetry peak at 3 D downstream switches to the right-hand side as shown in the experimental data. This location-switch phenomenon is also successfully captured by the present study. For the TKE profiles, as shown in Fig. 9(d), the present result matches the measurement in trend. However, an obvious difference is observed in the range of $0.25 < z/R < 0.75$ in which the TKE levels are 1 order of magnitude lower than the measurement. Comparing the measured data at 3 D downstream to the data at 1 D downstream, it suggest that the TKE in that region increases much faster in the experiment than it is in the simulation. A similar trend is found in the other two results obtained also from RANS calculations. However, in the two LES calculations, the TKE profiles at 3 D downstream match the measured data well. This is, again, interesting, because both LES simulations failed to capture the asymmetry in the deficit profiles, while successfully reproducing the TKE levels. At 3 D downstream, the asymmetry peaks of the deficit

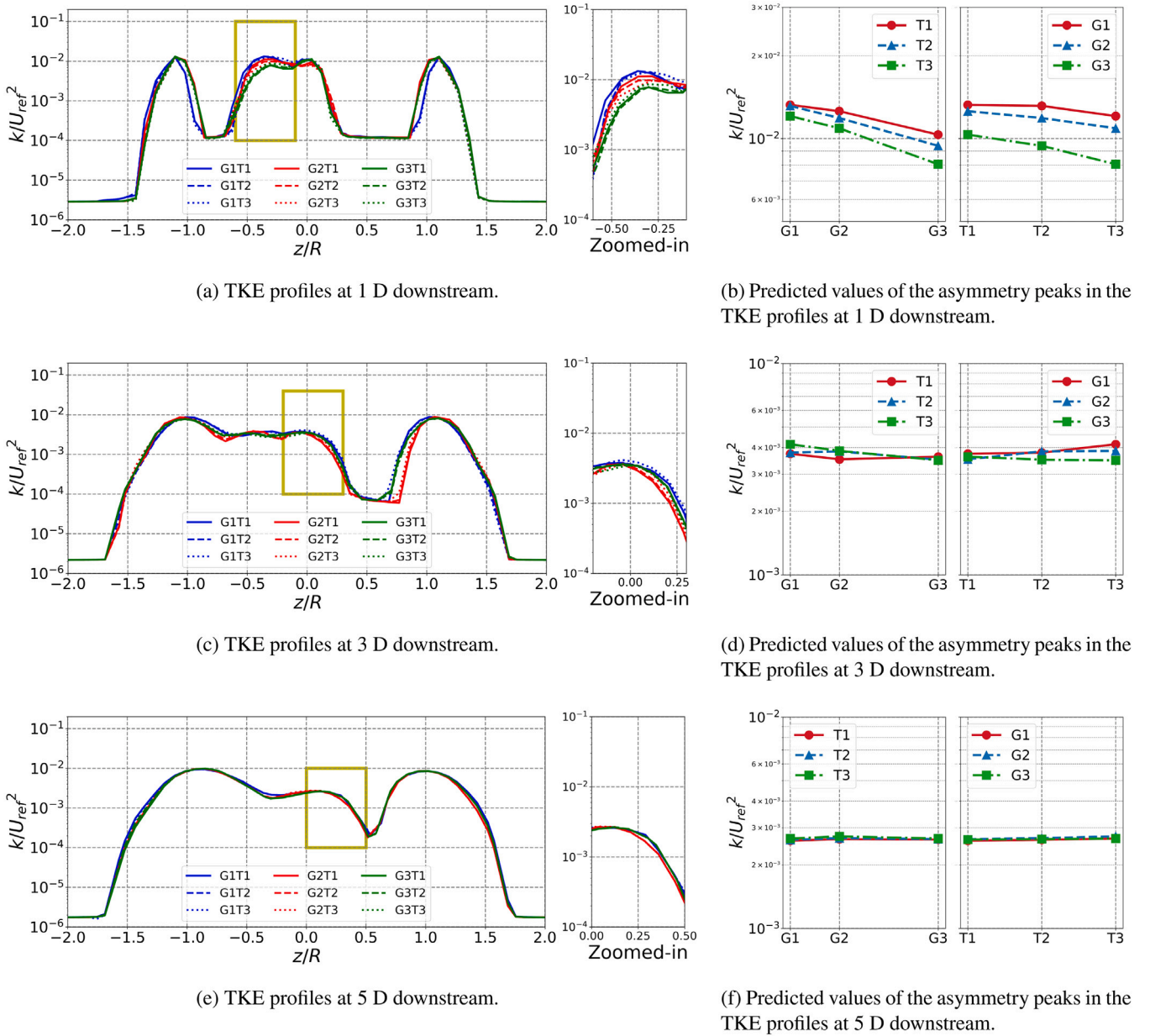


Fig. 7. TKE profiles obtained from all simulations.

and TKE profiles measured in the experiment are within the estimated numerical uncertainty ranges, thus the predicted results are validated.

At 5 D downstream, as shown in Fig. 9(e), the present results match the measured data well. Again, the present result is the only one successfully captured the asymmetry in the deficit profile. For the TKE profiles, as shown in Fig. 9(f), the present result matches the measured data in trend. However, an obvious difference still exist in the range of $0.25 < z/R < 0.75$ in which the TKE level is significantly under-predicted. The profile obtained by the AL-IB-LES method, on the other hand, match the measured TKE profile perfectly while failed to capture the asymmetry in the deficit profile. At 5 D downstream, the asymmetry peaks of the deficit and TKE profiles in the measurement are within the estimated numerical uncertainty ranges, thus the validation of the numerical results is achieved.

By comparing the profiles obtained from CFD simulations, i.e. RANS and LES, against the experimental data, it suggests that the difference between the present results and the measurement in the TKE profiles may be due to the limitations of the turbulence model used in the

present study. The 2-equation turbulence models are based on the Boussinesq hypothesis and the production of the turbulence kinetic energy is assumed to be isotropic. However, due to the rotation of the wind turbine, strong non-isotropy in the turbulence is present thus the 2-equation models may not be able to accurately predict the TKE levels in the wind turbine wake. However, it should also be concluded that even though in some particular regions the TKE levels are under-predicted, the overall profiles obtained by the present study match the experimental data well.

5. Flow details

In this section, the instantaneous flow fields are visualized to elucidate the underlying mechanism leading to the observations in the wake profiles, i.e. the asymmetry in the deficit profiles and the location-switch of the asymmetry peaks. The flow field obtained from the G1T1 case will be adopted in the following discussions.

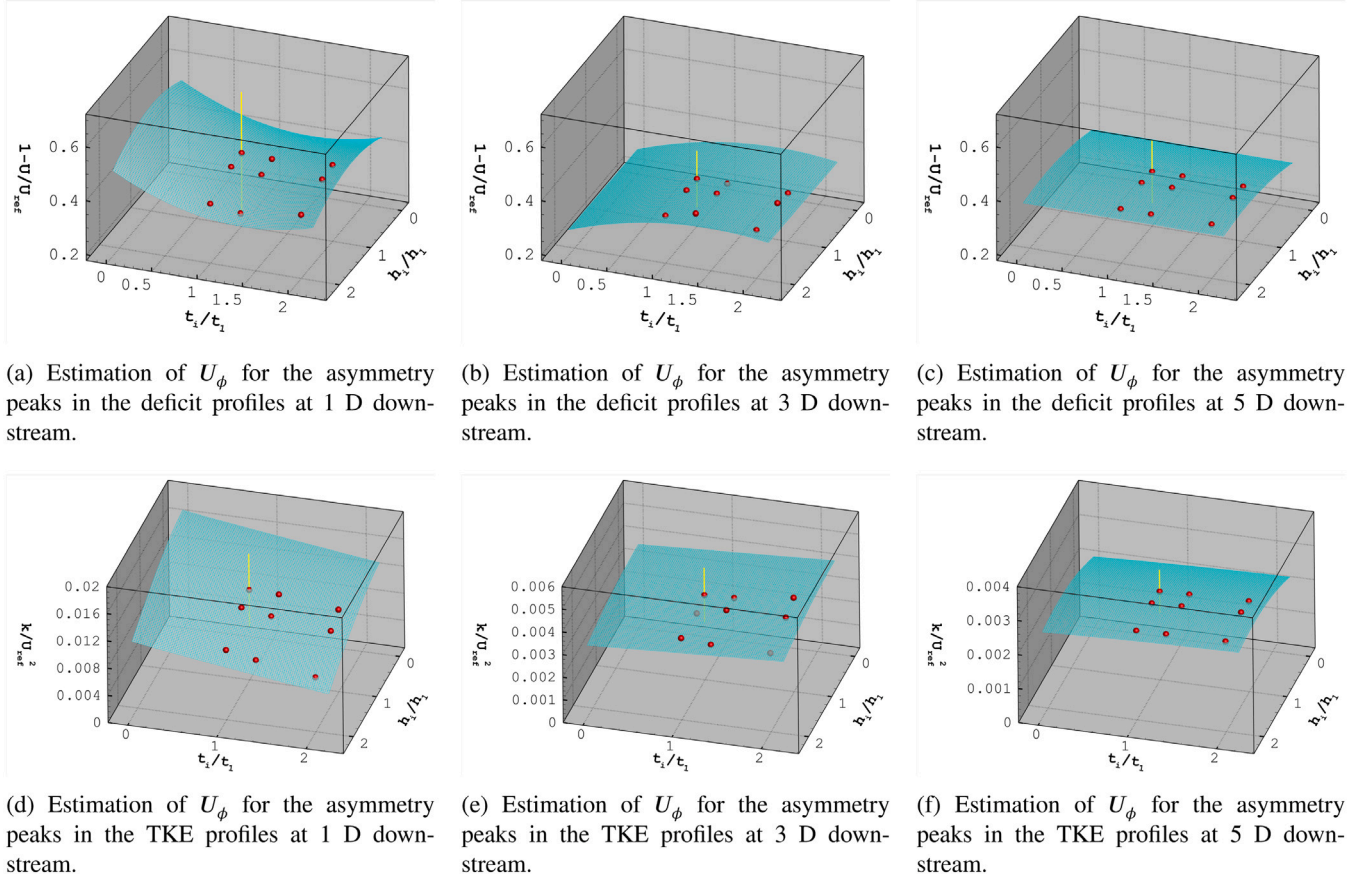


Fig. 8. Discretization uncertainty (U_ϕ) estimations of the mean wake deficit and mean TKE profiles for case GIT1. The yellow lines represent the discretization uncertainties.

To provide a general idea of the obtained wind turbine wake, the streamwise velocity contours on the xOy -plane and the vortex structures of the wind turbine wakes are visualized. First, it can be seen in Fig. 10 that an induction region, characterized by a low velocity region, is formed in front of the turbine. In the wake region, the flow velocity is lower than the free stream velocity, and complicated flow structures exist. Then, as shown in Fig. 11, iso-surfaces of Q value [61,62] equals 1000 are presented to show the wake structures. Note that the Q iso-surfaces are colored by streamwise velocity. It can be seen from the figure that the helical blade tip vortex, the blade root/nacelle vortex, and the tower vortex are clearly resolved. The three-dimensional vortex structures provide a good general idea of the wind turbine wakes, however, the details of the interactions between the vortex structures, i.e. the blade-generated wakes (referred to as blade wakes hereafter) and the tower wake, can be hardly identified in this view. Therefore, we will take advantage of the cross-sectional views of the computational domain in the following discussions.

5.1. Skew of the tower wake

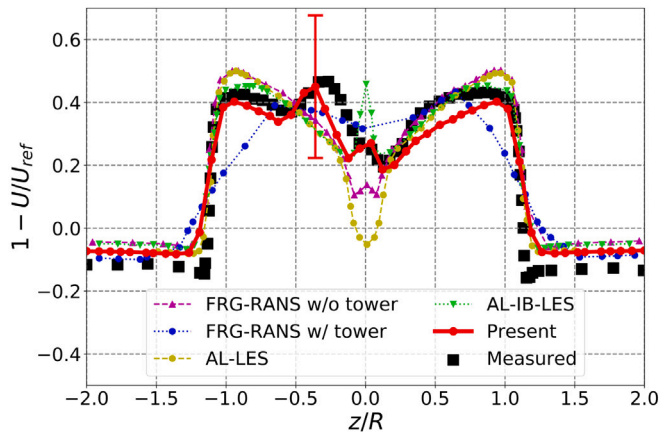
According to Newton's third law of motion, there must be an equal and opposite reaction for every action. In the case of a rotating wind turbine, the flow applies a torque on the wind turbine and makes it rotate, and thus conversely the wind turbine must impart an opposite torque on the flow. Therefore, the flow behind a wind turbine rotates in the opposite direction to that turbine [3,63]. In the current work, as mentioned earlier, the turbine is rotating in the negative direction about the x -axis. Therefore, as a result of the conservation of angular momentum and the fact that the wind turbine experiences a positive torque at $TSR = 6$, the blade wakes are forced to rotate in the positive direction about the x -axis as opposed to the wind turbine.

In the case of flow passing a circular cylinder, the vortices shed from the cylinder should follow the streamwise direction. However, in the current study, due to the rotation of the blade wakes, the incoming flow felt by the tower will no longer strictly point to the positive x -axis, but will have a transverse component that points to the negative z -axis. As a result, the tower wake is skewed and its propagation will have a directional preference.

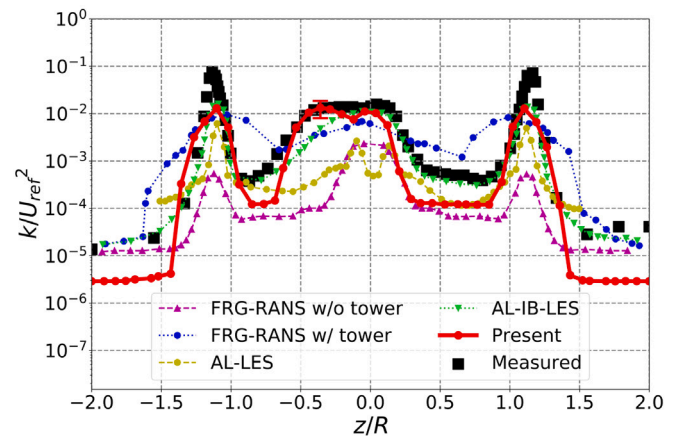
Fig. 12 shows the skewed tower wake at three different vertical locations (y -axis). The streamwise velocity contours are plotted on three y -normal planes, i.e. $y = -0.1$ m, $y = -0.2$ m, and $y = -0.3$ m, and the center line of each plane is indicated by a black dash-dotted line. For a better understanding of the wake structure, the streamwise velocity contours on the x -normal plane at 3 D downstream ($x = 2.682$ m) are also visualized. In Fig. 12, the color of the tower wake is blue and green, indicating its velocity is slower than the blade wakes. It can be clearly seen that the propagation of the tower wake has a directional preference, and this phenomenon will break the symmetry of the downstream wake profiles.

5.2. Asymmetry of the wake profiles

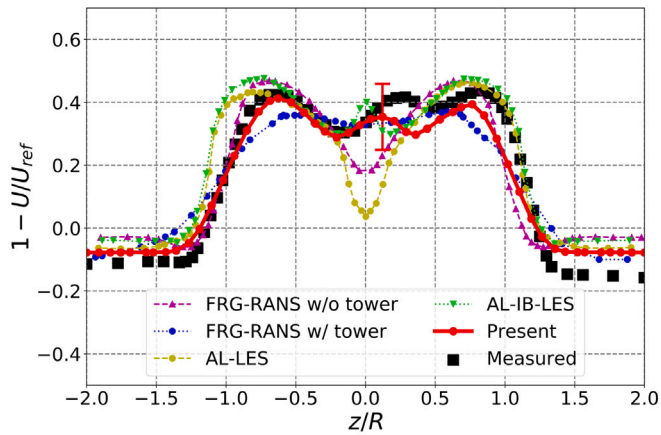
The instantaneous flow field is shown in Fig. 13. For the left-hand side figures, the instantaneous streamwise velocity contours at the 25th revolution are plotted on the x -normal planes at different downstream locations. White dashed lines indicate the hub height locations, i.e. $y = 0$. The black arrows represent the velocity vectors on the x -normal planes. In the right-hand side figures, the instantaneous profiles at the hub height are shown as the red solid lines, and the envelopes of the instantaneous profiles are shown as the gray bands. The measured and predicted mean deficit profiles are also plotted. All the figures are shown as if the observer is looking toward the positive direction of the



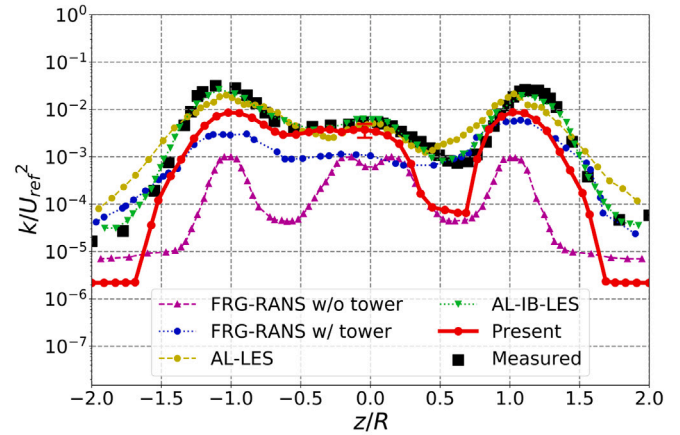
(a) Deficit profiles at 1 D downstream.



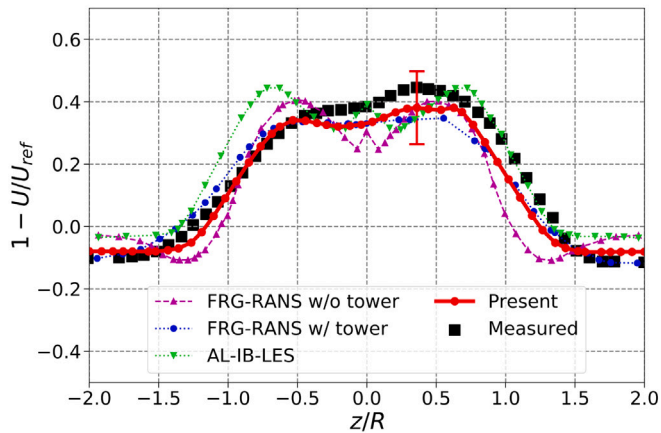
(b) TKE profiles at 1 D downstream.



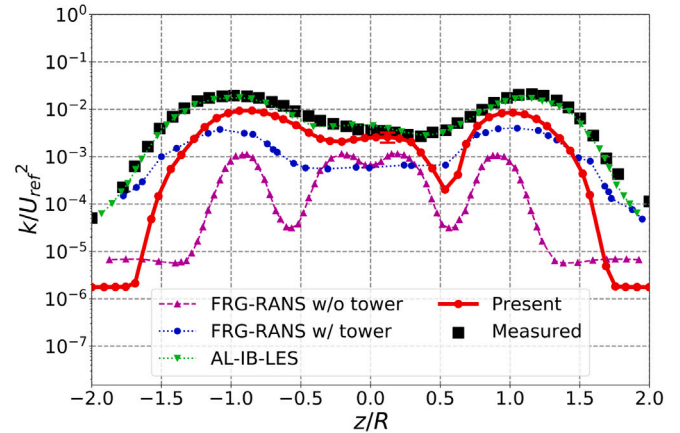
(c) Deficit profiles at 3 D downstream.



(d) TKE profiles at 3 D downstream.



(e) Deficit profiles at 5 D downstream.



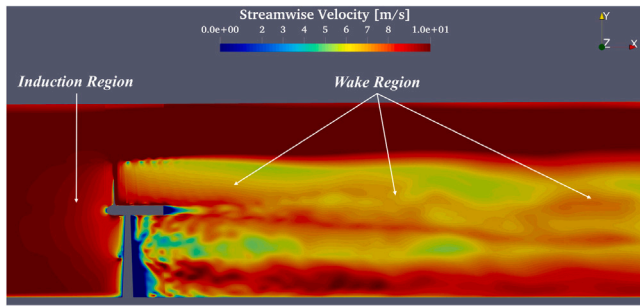
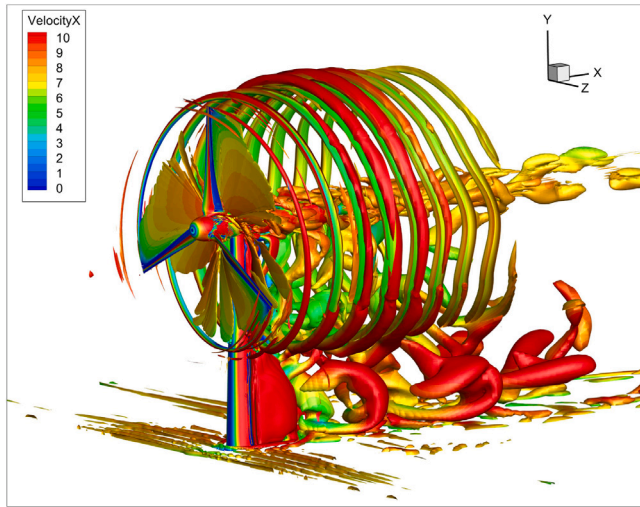
(f) TKE profiles at 5 D downstream.

Fig. 9. Comparison of mean wake deficit and TKE profiles.

x – axis. Therefore, the wind turbine rotates in the counter-clockwise direction and the blade wakes rotate in the clockwise direction in Figs. 13(a), 13(c), and 13(e).

At 1 D downstream, as shown in Fig. 13(a), a clear wake region is captured. The edge between the turbine wake and the free stream can be clearly identified. It can be observed that the tower wake is slower compared to the blade wakes and is on the left of the tower. Although

the tower is below the hub, the tower wake is brought to the hub height by the rotating blade wakes. Therefore, a peak in the deficit profile is expected in the corresponding location. As shown in Fig. 13(b), an asymmetry peak is shown at $z/R = -0.25$, indicating that the velocity at the hub height is significantly affected by the tower wake. In addition, the envelope of the instantaneous profiles has a wider width in the range of $-0.75 < z/R < 0.25$. Compared to the narrow envelope in

Fig. 10. Streamwise velocity contours on the xOy -plane.Fig. 11. Iso-surfaces of $Q = 1000$, colored by streamwise velocity.

the range of $0.25 < z/R < 1.0$, it shows that the flow is highly unsteady in the range of $-0.75 < z/R < 0.25$ due to the influence of the tower wake.

At 3 D downstream, the wake characteristics changes. The tower wake is carried by the blade wakes to a higher position above the hub height, and its shape is also distorted. Part of the tower wake has merged with the tip wakes and cannot be distinguished. For the deficit profiles, as shown in Fig. 13(d), comparing the curves in the ranges of $-1.25 < z/R < -0.75$ and $0.75 < z/R < 1.25$, the latter one has a steeper slope than the former. This indicates an enhanced mixing of the turbulence due to the presence of the tower wake on the negative z -axis, and thus the momentum inside and outside the wake region exchanges more efficiently. As a result, the velocity deficit in that region will be smoothed sooner as the downstream distance increases compared to that on the positive z -axis. Further, the asymmetry peaks are still observed while its location switches to the positive z -axis. This location-switch phenomenon can be easily understood by looking at the velocity contour plot, i.e. Fig. 13(c), in which the tower wake is carried and distorted by the rotating turbine wake.

Finally, Fig. 13(e) shows the velocity contours at $x = 5$ D. The tower wake is still visible at 5 D downstream. By comparing to Fig. 13(c), it is clear that the tower wake is rotating in the clockwise direction. For the deficit profiles, the slopes of the profiles around the tip locations are further decreased and the profiles are overall smoother than the profiles at 3 D downstream. Although not as apparent as it is at the previous two downstream locations, the effect of the tower wake is still visible and is characterized as the asymmetry peak at around $z/R = 0.25$. The location of the asymmetry peak is in line with the location of the tower wake shown in Fig. 13(e).

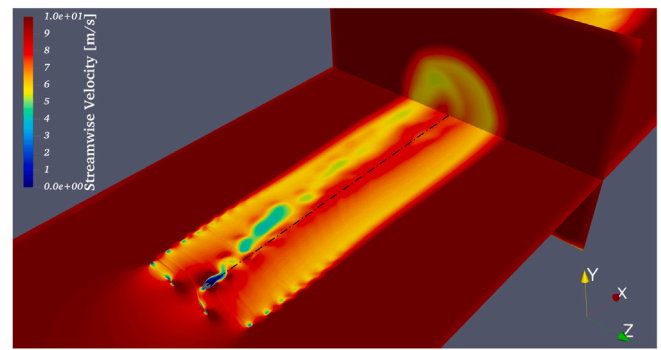
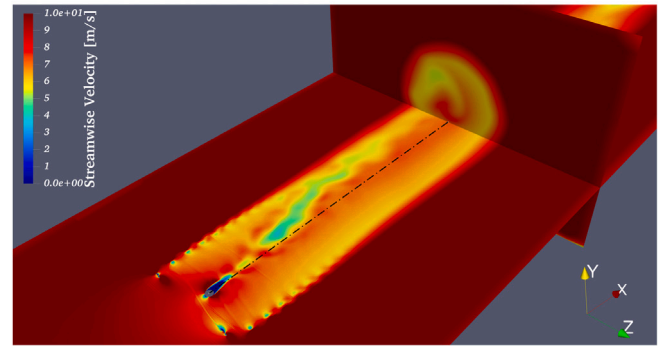
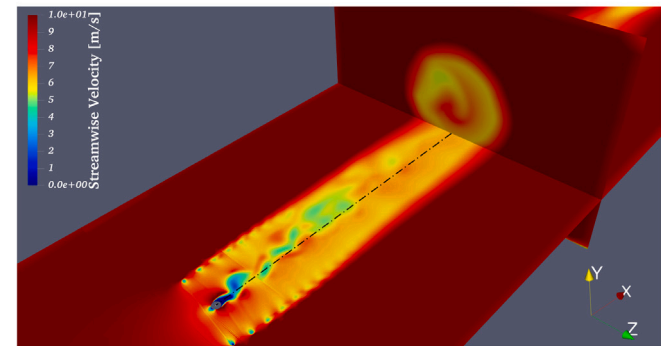
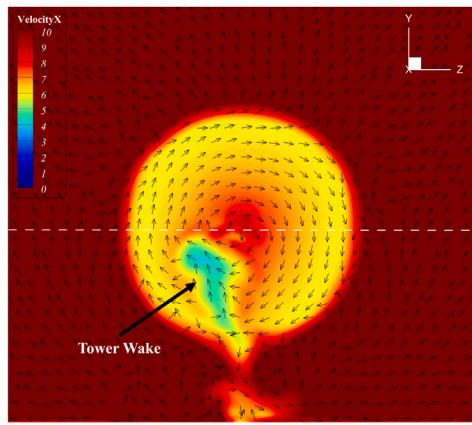
(a) Velocity contours at $y = -0.1$ m.(b) Velocity contours at $y = -0.2$ m.(c) Velocity contours at $y = -0.3$ m.

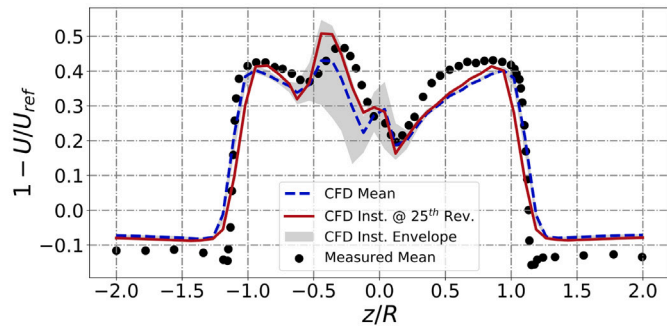
Fig. 12. Skewed tower wake at different vertical locations.

From the above observations and discussions, following conclusions can be drawn.

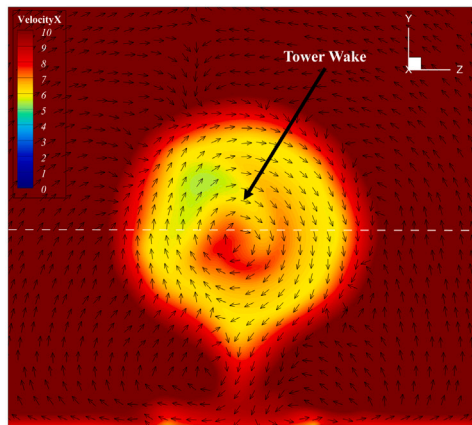
- The asymmetry of the deficit profiles is caused by the tower wake. Although the tower is below the hub height ($y < 0$), the tower wake is lifted to upper positions by the rotating blade wakes. As a result, the deficit profiles at the hub height are significantly affected by the slow tower wake. This tower effect is characterized as the asymmetry peaks in the deficit profiles.
- The location-switch phenomenon of the asymmetry peaks is owing to the rotation of the blade wakes. The tower wake is carried by the blade wakes and rotates in the clockwise direction. At 1 D downstream, the tower wake is propagating to the left. However, due to the rotation of the blade wakes, the tower wake is distorted and the position of the wake is changing as the downstream distance increases. As a result, the position of the asymmetry peak changes as the downstream distance increases.
- It is also worth pointing out that although the position of the asymmetry peak changes with downstream distance, at a fixed



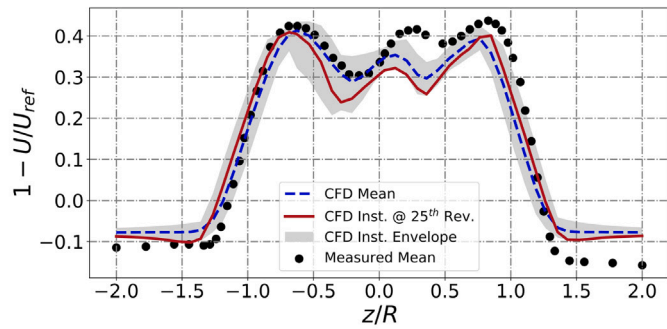
(a) Velocity contours at 1 D downstream.



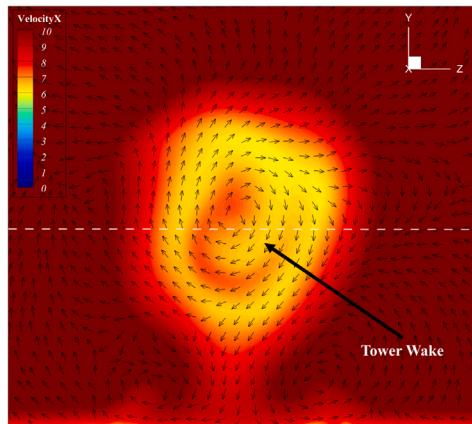
(b) Instantaneous deficit profiles at 1 D downstream.



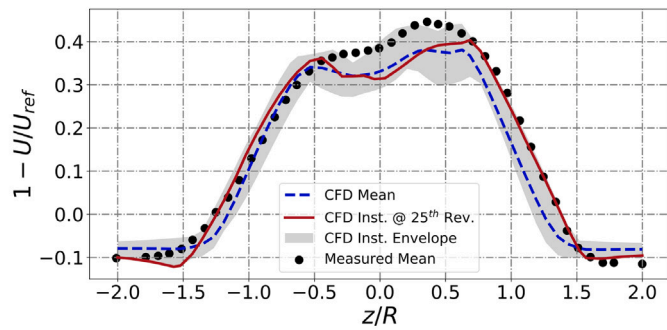
(c) Velocity contours at 3 D downstream.



(d) Instantaneous deficit profiles at 3 D downstream.



(e) Velocity contours at 5 D downstream.



(f) Instantaneous deficit profiles at 5 D downstream.

Fig. 13. Instantaneous wake contours and profiles.

downstream location, the position of the peak does not vary significantly over time. As discussed earlier, the rotation of the blade wakes is the driving force that makes the tower wake skew, and eventually breaks the symmetry in the downstream wake profiles. Therefore, because the turbine blades are rotating with a constant angular velocity, the blade wakes should also rotate at a steady speed. As a result, the path along which the tower wake propagates is relatively stable and will not vary significantly with time. This can be proved by the envelopes of the instantaneous wake profiles as shown in Figs. 13(b), 13(d), and 13(f). It can be

observed that although the magnitudes of the asymmetry peaks are changing over time, the position of which at each of the fixed downstream distances does not vary evidently.

- We confirm that the asymmetry in the wake profiles is physical and it is not a measurement error as discussed and suspected in some previous studies [19].
- We demonstrated that the RANS equations coupled with the $k-\omega$ SST turbulence model adopted in the current study is capable of capturing the asymmetry in the wake characteristics of the BT1 turbine.

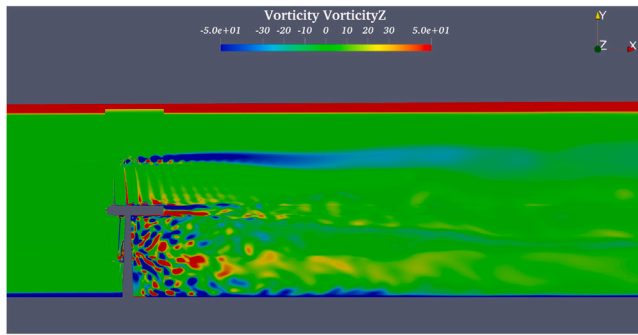


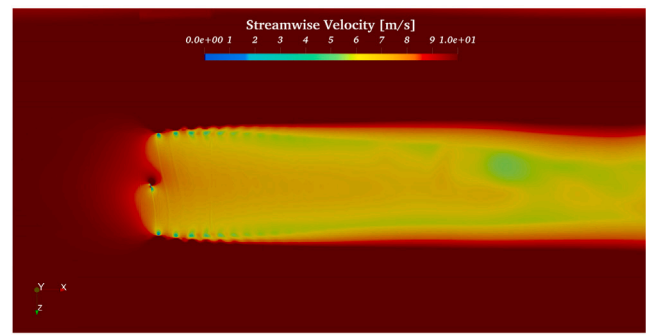
Fig. 14. Z-Vorticity contours on the xOy -planes.

5.3. Wake interactions

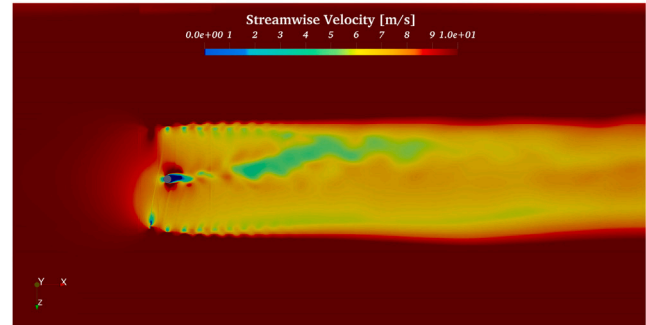
The wake structure of the wind turbine on the center z -normal plane (xOy -plane) is shown in Fig. 14. In general, three different flow regions can be identified from this figure. For the flow region above the hub height, i.e. $y > 0$, a steady and clear blade vortex structure can be observed. The blade wakes are visible up to roughly $1.5 D$ downstream, and then gradually distorted and dissipated. However, for the flow region below the hub height, i.e. $y < 0$, a highly unsteady and disordered flow pattern is presented compared to the flow pattern above the hub height. The blade wakes are not distinguishable in this wake region. This is a result of the interaction between the blade wakes and the tower wake. As can be observed in the figure, the highly complicated vortex structures are observed behind the tower. Further, for the flow region above the ground but below the lowest position of the blade tips ($ground < y < -R$), the vortex structure is solely generated by the tower wake, i.e. no blade wakes, but is only visible in the near wake region. As the downstream distance increases, due to the three dimensional nature of turbulence, that tower vortex is distorted by the vortices from upper regions and soon becomes highly unsteady and eventually dissipated.

The above discussions are more transparent by leveraging the cross-sectional slices of the flow field at different y locations. Fig. 15 shows the velocity contours at three selected y locations, i.e. $y = +0.5R$, $y = -0.5R$, and $y = -1.5R$, which respectively represent the wake structure consisting of solely the blade wakes, the blade wakes and the tower wake, and solely the tower wake. In Fig. 15(a), it can be observed that the blade wakes remain axisymmetric up to about $1.5 D$ downstream. As the downstream distance increases, as discussed earlier, the tower wake is lifted by the rotating blade wakes, and thus even for the flow region above the hub height, i.e. $y > 0$, interaction between the blade wakes and the tower wake will occur. At $y = -0.5R$, as shown in Fig. 15(b), the interaction between the blade wakes and the tower wake can be clearly seen, and the propagation of the tower wake has an obvious directional preference, as discussed in the previous sections. However, at $y = -1.5R$ as shown in Fig. 15(c), the tower wake propagates along the streamwise direction, and no evident directional preference is observed. This is because the tower wake in this flow region is away from the rotating blade wakes, and thus the propagation direction of it will be less affected. These distinctions among the flow structures in the three different flow regions reveal the importance of resolving the tower structure for accurate wind turbine simulations.

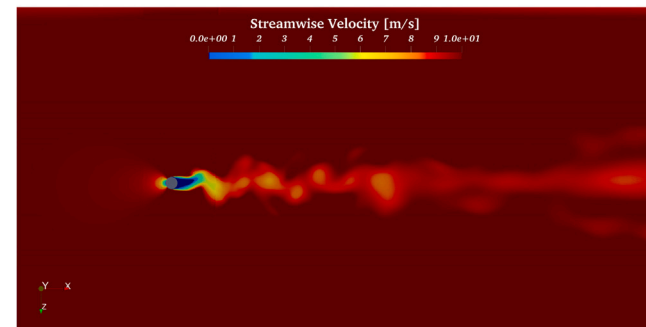
Lastly, Fig. 16 shows the deficit profiles along a vertical line on the xOy -plane at $1 D$ downstream. Note that the measured data is reported by Mittal et al. [42] and was not presented in the original report [56]. The numerical results obtained by Mittal et al. [42] is also presented here for comparison purpose. It can be observed that for $y/R > 0$, the envelope of the instantaneous wake profiles is invisible, implying that the streamwise velocity in that region is almost steady. However, for $y/R < 0$, that envelope is broad, meaning that the streamwise velocity is highly unsteady due to the interactions between the blade and the tower wakes.



(a) Velocity contours at $y = +0.5R$.



(b) Velocity contours at $y = -0.5R$.



(c) Velocity contours at $y = -1.5R$.

Fig. 15. Comparison of velocity contours at different y locations.

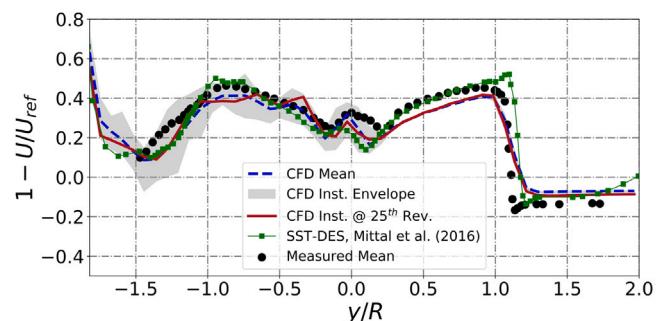


Fig. 16. Deficit profiles along a vertical line at $1 D$ downstream.

6. Conclusions

In this study, unsteady RANS calculations using fully resolved turbine geometries were performed aiming at the wake characteristics of the NTNU BT1 wind turbine. The MVG approach together with

sliding interfaces is leveraged to handle the relative motion between the rotating turbine and the stationary tower.

Before examining the CFD-predicted wake characteristics, efforts had been made to demonstrate the credibility of the obtained simulation results. First, a systematic verification procedure was performed to estimate the numerical uncertainties of the predicted wake characteristics. The asymmetry peaks in the deficit profiles were identified and selected as the metric to quantify the discretization uncertainties. It was found that the maximum discretization uncertainties for both the deficit and TKE profiles appeared at 1 D downstream, and then decreased with the increase of the downstream distance. A validation study was also performed by comparing the CFD predictions against the experimental data. In general, good agreement between the prediction and the measurement was achieved and the numerical results were validated.

Then, the predictions were compared to the results obtained by 4 other representative modeling approaches to demonstrate the capabilities and limitations of the methods adopted in the current work. It was shown that the asymmetry in the deficit profiles was successfully captured in the present study, whereas all other approaches predicted symmetric deficit profiles. However, it was observed that the present results significantly under-predicted the TKE levels in certain regions, and it was suspected that the discrepancies between the present predictions and the measured data are owing to the limitations of the 2-equation turbulence models which assume the production of TKE to be isotropic.

Afterward, the instantaneous flow fields were visualized and discussed to elucidate the underlying mechanism which leads to the asymmetric wake profiles. It was identified that the rotation of the blade wakes is the driving force that makes the tower wake skew. As a result, the propagation of the tower wake had an obvious directional preference. Although the tower was below the hub height, the tower wake was carried upwards by the rotating blade wakes, and eventually broke the symmetry of the wake profiles in the upper regions.

The location-switch phenomenon of the asymmetry peaks was demonstrated to be another result of the rotation of the blade wakes. It was shown that the tower wake was forced to rotate by the blade wakes, and its position was changing with downstream distance as it was propagating. As a result, the position of the asymmetry peak switched from the negative z - axis to the positive side as the downstream distance increased from 1 D to 3 D. It was also discussed that although the position of the asymmetry peak differs at different downstream distances, it should not vary significantly at a fixed downstream location due to the fact that the wind turbine rotates at a constant speed.

Last, the interaction between the blade wakes and the tower wake was visualized. It was observed that above the hub height, the blade wakes formed a relatively steady flow pattern within the downstream distance of 1.5 D. However, with the increase of the downstream distance, the interaction between the blade wakes and the tower wake was also observed in the upper regions. For the region below the hub height but above the lowest position of the blade tips, obvious interaction was shown and the tower wake was skewed. For the region below the lowest position of the blade tips, however, tower wake propagating along the free stream direction was observed, which indicated that the tower wake was less affected by the rotating blade wakes in this region.

From the results and discussions, we highlight the importance of resolving the tower structure to the accurate prediction of the wake characteristics in wind turbine simulations. Although higher-fidelity CFD simulations, i.e. PANS and LES, were applied in prior studies in an attempt to capture the asymmetric wake profiles of the NTNU BT1 wind turbine, the underlying mechanism was not fully understood due to the absence of tower structure. We confirmed that the asymmetry in the wake profiles is physical and is not a measurement error as suspected in some prior studies. We also demonstrated that the RANS equations coupled with the $k-\omega$ SST turbulence model are capable of simulating the wake characteristics of the NTNU BT1 wind turbine.

CRediT authorship contribution statement

Maokun Ye: Conceptualization, Methodology, Formal analysis, Writing – original draft. **Hamn-Ching Chen:** Supervision, Conceptualization, Data curation, Writing – review & editing. **Arjen Koop:** Resources, Data curation, Writing – review & editing.

Data availability

Data will be made available on request.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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